

Volume Iii Bounds Sequences And Series Reference

Contents

0.1	Real Line One Dimensional	4
0.2	Order Distance Length	4
0.3	Absolute Value as Distance	4
0.4	Intervals as Subsets	4
0.5	Neighborhoods and Balls	5
0.6	Open Sets	5
0.7	Closed Sets	5
0.8	Interior Exterior Boundary	5
0.9	Limit and Isolated Points	5
0.10	Unions and Intersections	6
0.11	Covers and Subcovers	6
0.12	Compactness on $\mathbb{R}$	6
0.13	Heine Borel	6
0.14	Completeness Axiom	7
0.14.1	Nested Intervals	7
0.14.2	Archimedean Property	7
0.14.3	Density	7
0.14.4	Completeness Summary	8
0.15	Bounds and Extremal Values	8
0.15.1	Upper and Lower Bounds	8
0.15.2	Suprema and Infima	8
0.15.3	Maxima and Minima	9
0.15.4	Epsilon Characterization	10
0.15.5	Extremal Bounds Summary	10
0.16	Algebra of Bounds	10
0.16.1	Set Operations and Bounds	10
0.16.2	Lattice Operations and Bounds	11
0.16.3	Images, Inverse Images, and Extrema	11
0.16.4	Dilation and Displacement	12
0.16.5	Algebra of Suprema and Infima	13
0.17	Algebra of Functions	16
0.18	Real-Valued Functions	16
0.19	Subsets of $\mathbb{R}$	18
0.20	Motivation	20
0.21	Foundations	20
0.21.1	Foundations	20
0.21.2	Summation Notation	20

0.21.3	The Binomial Theorem	20
0.22	Algebra of Summations	21
0.23	Difference Operators	21
0.23.1	Difference Operators	21
0.23.2	Higher Differences	21
0.23.3	Properties of the Difference Operator	22
0.23.4	Difference Tables and Polynomial Detection	22
0.24	Telescoping Sums	22
0.24.1	Telescoping Sums	22
0.25	Discrete Antiderivatives	22
0.25.1	Discrete Antiderivatives	22
0.26	Polynomial Basis	24
0.26.1	Polynomial Basis: Falling Factorials	24
0.27	Calculus Rules	24
0.27.1	Calculus Rules	24
0.28	Expansions	24
0.28.1	Newton's Forward Difference Expansion	24
0.29	Special Functions	24
0.29.1	Special Functions in Discrete Calculus	24
0.29.2	The Function 2^n in Discrete Calculus	25
0.30	Sequence Definitions	26
0.30.1	Definitions	26
0.30.2	Null and Constant Sequences	26
0.31	Examples and Counterexamples	28
0.32	Convergence	28
0.32.1	Limits	28
0.32.2	Algebra of Limits	29
0.33	Tails	31
0.34	Monotonicity	32
0.35	Subsequences	33
0.36	Divergence	35
0.37	Liminf and Limsup	36
0.38	Order-Theoretic Limit Constructions	37
0.39	Cluster Values of Sequences	38
0.40	Cauchy	38
0.41	Applications of Sequences	40
0.42	Finite Series	41
0.43	Infinite Series	44

Structure of the Real Line

0.1 Real Line One Dimensional

0.2 Order Distance Length

Definition 0.2.1 (Distance on the Real Line). The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Definition 0.2.2 (Length of a Bounded Interval). Let $a, b \in \mathbb{R}$ with $a \leq b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Theorem 0.2.3 (The Distance Function Is a Metric). *The map $d(x, y) = |x - y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \geq 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \leq d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space. Go to proof.*

0.3 Absolute Value as Distance

Proposition 0.3.1 (Absolute Value Is Distance to the Origin). *For every $a \in \mathbb{R}$, $|a| = d(a, 0)$. Go to proof.*

0.4 Intervals as Subsets

Definition 0.4.1 (Bounded Subset of the Real Line). A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \geq 0$ such that $|x| \leq M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Set Operations on Intervals

Definition 0.4.2 (Union of Intervals). Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Definition 0.4.3 (Intersection of Intervals). Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Definition 0.4.4 (Complement of an Interval). Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Definition 0.4.5 (Difference of Intervals). Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

0.5 Neighborhoods and Balls

Definition 0.5.1 (Open Ball on the Real Line). Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{y \in \mathbb{R} : |y - x| < r\} = (x - r, x + r).$$

Definition 0.5.2 (Neighborhood of a Point). A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

0.6 Open Sets

Definition 0.6.1 (Open Set in $\mathbb{R}$). A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U \exists r > 0 (B_r(x) \subseteq U).$$

Theorem 0.6.2 (Open Intervals Are Open). *Every open interval (a, b) , every open ray (a, ∞) and $(-\infty, b)$, and $\mathbb{R}$ itself are open sets. Go to proof.*

Theorem 0.6.3 (Unions and Finite Intersections of Open Sets). *An arbitrary union of open subsets of $\mathbb{R}$ is open, and a finite intersection of open subsets of $\mathbb{R}$ is open. Go to proof.*

0.7 Closed Sets

Definition 0.7.1 (Closed Set in $\mathbb{R}$). A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Theorem 0.7.2 (Closed Sets Contain Their Limit Points). *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points. Go to proof.*

0.8 Interior Exterior Boundary

Definition 0.8.1 (Interior Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0 (B_r(x) \subseteq E).$$

Definition 0.8.2 (Exterior Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0 (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Definition 0.8.3 (Boundary Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0 (B_r(x) \cap E \neq \emptyset \wedge B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Definition 0.8.4 (Interior of a Set). The *interior* of $E \subseteq \mathbb{R}$, written $\text{int}(E)$, is the set of all interior points of E .

0.9 Limit and Isolated Points

Definition 0.9.1 (Limit Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0 \exists y \in E (0 < |y - x| < r).$$

Definition 0.9.2 (Isolated Point). Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0 \forall y \in E (|y - x| < r \Rightarrow y = x).$$

0.10 Unions and Intersections

Theorem 0.10.1 (Closure Laws for Closed Sets). *An arbitrary intersection of closed subsets of $\mathbb{R}$ is closed, and a finite union of closed subsets of $\mathbb{R}$ is closed. Go to proof.*

0.11 Covers and Subcovers

Definition 0.11.1 (Open Cover). Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i \in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i \in I} U_i.$$

Definition 0.11.2 (Subcover and Finite Subcover). Let $\{U_i\}_{i \in I}$ be an open cover of E . A *subcover* is a subfamily $\{U_i\}_{i \in J}$ with $J \subseteq I$ that still covers E . It is a *finite subcover* when J is finite.

0.12 Compactness on $\mathbb{R}$

Definition 0.12.1 (Compact Set in $\mathbb{R}$). A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i \in I} \text{ open cover of } K \exists \text{finite } J \subseteq I \left(K \subseteq \bigcup_{i \in J} U_i \right).$$

Theorem 0.12.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded). *If $K \subseteq \mathbb{R}$ is compact, then K is closed and bounded. Go to proof.*

0.13 Heine Borel

Theorem 0.13.1 (Closed Bounded Intervals Are Compact). *Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact. Go to proof.*

Theorem 0.13.2 (Heine–Borel Theorem for $\mathbb{R}$). *A set $K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded. Go to proof.*

Proofs

Bounds

0.14 Completeness Axiom

Axiom of Completeness

Definition 0.14.1 (Least Upper Bound Property). Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S .

0.14.1 Nested Intervals

Nested Interval Property

Theorem 0.14.2 (Nested Interval Property). Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Go to proof.

0.14.2 Archimedean Property

Archimedean Property

Theorem 0.14.3 (Archimedean Property). For every real number x , there exists a natural number n such that $n > x$.

Go to proof.

Corollary 0.14.4 (Archimedean Reciprocal Corollary). For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.

Go to proof.

0.14.3 Density

Density

Definition 0.14.5 (Dense Subset). Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in S* if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Definition 0.14.6 (Order Dense Subset). Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in X* if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Theorem 0.14.7 (Density of the Rational Numbers in $\mathbb{R}$). For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$. *Go to proof.*

Theorem 0.14.8 (Density of the Irrational Numbers in $\mathbb{R}$). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$. Go to proof.*

Corollary 0.14.9 (Rational and Irrational Points in Every Open Interval). *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number. Go to proof.*

0.14.4 Completeness Summary

0.15 Bounds and Extremal Values

0.15.1 Upper and Lower Bounds

Definition 0.15.1 (Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $b \in S$. A *bound* of A means a one-sided comparison between b and every element of A : either $x \leq b$ for all $x \in A$, or $b \leq x$ for all $x \in A$.

Definition 0.15.2 (Upper Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $u \in S$. The element u is an *upper bound* of A if every element of A is less than or equal to u .

Definition 0.15.3 (Lower Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $\ell \in S$. The element ℓ is a *lower bound* of A if every element of A is greater than or equal to ℓ .

Definition 0.15.4 (Bounded Above). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded above* if it has at least one upper bound in S .

Definition 0.15.5 (Bounded Below). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded below* if it has at least one lower bound in S .

Definition 0.15.6 (Bounded Set). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded* if it is bounded above and bounded below.

0.15.2 Suprema and Infima

Definition 0.15.7 (Supremum). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $s \in S$. The element s is the *supremum* of A if s is an upper bound of A and no smaller element of S is an upper bound of A .

Definition 0.15.8 (Infimum). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $i \in S$. The element i is the *infimum* of A if i is a lower bound of A and no larger element of S is a lower bound of A .

Proposition 0.15.9 (Uniqueness of the Supremum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s, t \in S$ are both suprema of A , then $s = t$.*

Go to proof.

Proposition 0.15.10 (Uniqueness of the Infimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i, j \in S$ are both infima of A , then $i = j$.*

Proposition 0.15.11 (Upper Bound Property of the Supremum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $s = \sup A$ exists in S . Then every element of A lies below s :*

$$(\forall x \in A)(x \leq s).$$

Go to proof.

Proposition 0.15.12 (Lower Bound Property of the Infimum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $i = \inf A$ exists in S . Then every element of A lies above i :*

$$(\forall x \in A)(i \leq x).$$

Lemma 0.15.13 (Subset Inclusion Preserves Upper Bounds). *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If u is an upper bound of B , then u is an upper bound of A .*

Go to proof.

Lemma 0.15.14 (Subset Inclusion Preserves Lower Bounds). *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If ℓ is a lower bound of B , then ℓ is a lower bound of A .*

Theorem 0.15.15 (Supremum Is Monotone Under Inclusion). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*

Go to proof.

Theorem 0.15.16 (Infimum Is Monotone Under Inclusion). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. If $A \subseteq B$, then $\inf B \leq \inf A$.*

Proposition 0.15.17 (Upper Bound Comparison with the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*

Go to proof.

Proposition 0.15.18 (Lower Bound Comparison with the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. For $\ell \in \mathbb{R}$, the number ℓ is a lower bound of A if and only if $\ell \leq i$.*

Proposition 0.15.19 (Every Element Lies Below the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*

Go to proof.

Proposition 0.15.20 (Every Element Lies Above the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. Then $i \leq x$ for every $x \in A$.*

Theorem 0.15.21 (Infimum Is Less Than or Equal to the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then $\inf A \leq \sup A$.*

Go to proof.

Proposition 0.15.22 (Order Comparison of Suprema). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If every $x \in A$ is below some $y \in B$, then $\sup A \leq \sup B$.*

Go to proof.

Theorem 0.15.23 (Least Upper Bound Property Implies Existence of Suprema). *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number s such that $s = \sup S$.*

Theorem 0.15.24 (Greatest Lower Bound Property Implies Existence of Infima). *Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Then there exists a real number i such that $i = \inf S$.*

0.15.3 Maxima and Minima

Definition 0.15.25 (Maximum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *maximum* of A if $m \in A$ and every element of A is less than or equal to m .*

Definition 0.15.26 (Minimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *minimum* of A if $m \in A$ and every element of A is greater than or equal to m .*

Proposition 0.15.27 (Uniqueness of the Maximum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both maxima of A , then $m = n$.*

Go to proof.

Proposition 0.15.28 (Uniqueness of the Minimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both minima of A , then $m = n$.*

Proposition 0.15.29 (Maximum Implies Supremum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \max A$, then $m = \sup A$.*

Go to proof.

Proposition 0.15.30 (Minimum Implies Infimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \min A$, then $m = \inf A$.*

Proposition 0.15.31 (Supremum in the Set is the Maximum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s = \sup A$ and $s \in A$, then $s = \max A$.*

Go to proof.

Proposition 0.15.32 (Infimum in the Set is the Minimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i = \inf A$ and $i \in A$, then $i = \min A$.*

0.15.4 Epsilon Characterization

ε -Characterization of Supremum

Definition 0.15.33 (Epsilon Characterization of Supremum). Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Definition 0.15.34 (Epsilon Characterization of Infimum). Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

0.15.5 Extremal Bounds Summary

0.16 Algebra of Bounds

0.16.1 Set Operations and Bounds

Lemma 0.16.1 (Union Preserves Upper Bounds). *Let $A, B \subseteq \mathbb{R}$. If u_A is an upper bound of A and u_B is an upper bound of B , then $\max\{u_A, u_B\}$ is an upper bound of $A \cup B$. Go to proof.*

Lemma 0.16.2 (Union Preserves Lower Bounds). *Let $A, B \subseteq \mathbb{R}$. If ℓ_A is a lower bound of A and ℓ_B is a lower bound of B , then $\min\{\ell_A, \ell_B\}$ is a lower bound of $A \cup B$. Go to proof.*

Proposition 0.16.3 (Union Is Bounded Above Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded above if and only if A is bounded above and B is bounded above. Go to proof.*

Proposition 0.16.4 (Union Is Bounded Below Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded below if and only if A is bounded below and B is bounded below. Go to proof.*

Corollary 0.16.5 (Union Is Bounded Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded if and only if A is bounded and B is bounded. Go to proof.*

Lemma 0.16.6 (Subsets Preserve Upper Bounds). *Let $C \subseteq A \subseteq \mathbb{R}$. If u is an upper bound of A , then u is an upper bound of C . Go to proof.*

Lemma 0.16.7 (Subsets Preserve Lower Bounds). *Let $C \subseteq A \subseteq \mathbb{R}$. If ℓ is a lower bound of A , then ℓ is a lower bound of C . Go to proof.*

Corollary 0.16.8 (Intersections Inherit Bounds). *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \cap B$, and every lower bound of A is a lower bound of $A \cap B$. Go to proof.*

Corollary 0.16.9 (Differences Inherit Bounds). *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \setminus B$, and every lower bound of A is a lower bound of $A \setminus B$. Go to proof.*

Corollary 0.16.10 (Complements Inherit Ambient Bounds). *Let $A \subseteq E \subseteq \mathbb{R}$, and take complements relative to E . Every upper bound of E is an upper bound of $E \setminus A$, and every lower bound of E is a lower bound of $E \setminus A$. Go to proof.*

0.16.2 Lattice Operations and Bounds

Definition 0.16.11 (Pointwise Maximum and Minimum Sets). Let $A, B \subseteq \mathbb{R}$ be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A, b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A, b \in B\}.$$

Theorem 0.16.12 (Supremum of the Pointwise Maximum Set). Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.13 (Infimum of the Pointwise Maximum Set). Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

Go to proof.

Theorem 0.16.14 (Supremum of the Pointwise Minimum Set). Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.15 (Infimum of the Pointwise Minimum Set). Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

Go to proof.

0.16.3 Images, Inverse Images, and Extrema

Theorem 0.16.16 (Increasing Image Preserves Suprema). Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \rightarrow \mathbb{R}$ be increasing. Suppose $s = \sup A$ belongs to I and f is continuous at s . Then

$$\sup f(A) = f(\sup A), \quad f(A) = \{f(a) : a \in A\}.$$

Go to proof.

Theorem 0.16.17 (Increasing Image Preserves Infima). Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \rightarrow \mathbb{R}$ be increasing. Suppose $t = \inf A$ belongs to I and f is continuous at t . Then

$$\inf f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.18 (Decreasing Image Sends Infimum to Supremum). Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \rightarrow \mathbb{R}$ be decreasing. Suppose $t = \inf A$ belongs to I and f is continuous at t . Then

$$\sup f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.19 (Decreasing Image Sends Supremum to Infimum). Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \rightarrow \mathbb{R}$ be decreasing. Suppose $s = \sup A$ belongs to I and f is continuous at s . Then

$$\inf f(A) = f(\sup A).$$

Go to proof.

Theorem 0.16.20 (Increasing Inverse Preserves Suprema). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \rightarrow J$ be bijective. Suppose $f^{-1} : J \rightarrow I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Theorem 0.16.21 (Increasing Inverse Preserves Infima). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \rightarrow J$ be bijective. Suppose $f^{-1} : J \rightarrow I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.22 (Decreasing Inverse Sends Infimum to Supremum). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \rightarrow J$ be bijective. Suppose $f^{-1} : J \rightarrow I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.23 (Decreasing Inverse Sends Supremum to Infimum). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \rightarrow J$ be bijective. Suppose $f^{-1} : J \rightarrow I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

0.16.4 Dilation and Displacement

Definition 0.16.24 (Displacement, Dilation, and Reflection). *Let $A \subseteq \mathbb{R}$ and let $c, \lambda \in \mathbb{R}$. Define*

$$A + c := \{a + c : a \in A\}, \quad A - c := \{a - c : a \in A\}, \quad \lambda A := \{\lambda a : a \in A\}.$$

The set $-A := (-1)A$ is the *reflection* of A through the origin.

Proposition 0.16.25 (Translation Preserves Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If u is an upper bound of A , then $u + c$ is an upper bound of $A + c$. Go to proof.*

Proposition 0.16.26 (Translation Preserves Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A , then $\ell + c$ is a lower bound of $A + c$. Go to proof.*

Proposition 0.16.27 (Positive Dilation Preserves Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If u is an upper bound of A , then λu is an upper bound of λA . Go to proof.*

Proposition 0.16.28 (Positive Dilation Preserves Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If ℓ is a lower bound of A , then $\lambda \ell$ is a lower bound of λA . Go to proof.*

Proposition 0.16.29 (Negative Dilation Sends Lower Bounds to Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If ℓ is a lower bound of A , then $\lambda \ell$ is an upper bound of λA . Go to proof.*

Proposition 0.16.30 (Negative Dilation Sends Upper Bounds to Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If u is an upper bound of A , then λu is a lower bound of λA . Go to proof.*

Corollary 0.16.31 (Reflection Swaps Upper and Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A , then $-\ell$ is an upper bound of $-A$. If u is an upper bound of A , then $-u$ is a lower bound of $-A$. Go to proof.*

0.16.5 Algebra of Suprema and Infima

Current Algebraic Properties of Supremum and Infimum

Definition 0.16.32 (Set Arithmetic Images). Let $A, B \subseteq \mathbb{R}$. Define

$$\begin{aligned} A + B &:= \{a + b : a \in A, b \in B\}, & A - B &:= \{a - b : a \in A, b \in B\}, \\ AB &:= \{ab : a \in A, b \in B\}, & A/B &:= \{a/b : a \in A, b \in B, b \neq 0\}. \end{aligned}$$

Definition 0.16.33 (Reciprocal Image). Let $A \subseteq \mathbb{R}$ and suppose $0 \notin A$. The reciprocal image of A is

$$A^{-1} := \{1/a : a \in A\}.$$

Theorem 0.16.34 (Translation Invariance of the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \quad A + c = \{a + c : a \in A\}.$$

Go to proof.

Theorem 0.16.35 (Translation Invariance of the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $c \in \mathbb{R}$. Then*

$$\inf(A + c) = \inf A + c.$$

Go to proof.

Theorem 0.16.36 (Scalar Multiplication by a Positive Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \quad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.37 (Positive Scalar Multiplication of the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k > 0$. Then*

$$\inf(kA) = k \inf A.$$

Go to proof.

Theorem 0.16.38 (Scalar Multiplication by a Negative Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \quad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.39 (Negative Scalar Multiplication of the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k < 0$. Then*

$$\inf(kA) = k \sup A.$$

Go to proof.

Theorem 0.16.40 (Negation Exchanges Supremum and Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Theorem 0.16.41 (Supremum of a Sum Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \quad A + B = \{a + b : a \in A, b \in B\}.$$

Go to proof.

Theorem 0.16.42 (Infimum of a Sum Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf(A + B) = \inf A + \inf B.$$

Go to proof.

Theorem 0.16.43 (Supremum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \quad A - B = \{a - b : a \in A, b \in B\}.$$

Go to proof.

Theorem 0.16.44 (Infimum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded below and B bounded above. Then*

$$\inf(A - B) = \inf A - \sup B.$$

Go to proof.

Theorem 0.16.45 (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \quad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Theorem 0.16.46 (Supremum of the Absolute-Value Image). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \quad |A| = \{|a| : a \in A\}.$$

Go to proof.

Theorem 0.16.47 (Supremum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

Go to proof.

Theorem 0.16.48 (Infimum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

Go to proof.

Theorem 0.16.49 (Supremum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.50 (Infimum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.51 (Supremum of a Quotient Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\sup(A/B) = \max \left\{ \frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B} \right\}.$$

Go to proof.

Theorem 0.16.52 (Infimum of a Quotient Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\inf(A/B) = \min \left\{ \frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B} \right\}.$$

Go to proof.

Proposition 0.16.53 (Bounds for Suprema and Infima Under Set Addition). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \quad \text{and} \quad \sup(A + B) \leq \sup A + \sup B.$$

Go to proof.

Proposition 0.16.54 (Upper Bounds Depend on the Ambient Order). *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order. Go to proof.*

Proposition 0.16.55 (Suprema Depend on the Ambient Set). *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S , if it exists, need not agree with the supremum of A relative to S' . Go to proof.*

Theorem 0.16.56 (Ambient Existence of Supremum). *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S . Go to proof.*

Lemma 0.16.57 (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. Go to proof.

Proofs

Capstone

Functions

0.17 Algebra of Functions

Algebra of Functions

Proposition 0.17.1 (Composition of Injections Is Injective). *Let $f : A \rightarrow B$ and let $g : B \rightarrow C$. If f is injective and g is injective, then $g \circ f : A \rightarrow C$ is injective.*

Go to proof.

Proposition 0.17.2 (Composition of Surjections Is Surjective). $\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \text{Surjective}(g \circ f)$.

Go to proof.

Corollary 0.17.3 (Composition of Bijections Is Bijective). *Let $f : A \rightarrow B$ and let $g : B \rightarrow C$. If f is bijective and g is bijective, then $g \circ f : A \rightarrow C$ is bijective.*

Go to proof.

Proposition 0.17.4 (Inverse of a Bijection Is a Bijection). $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$.

Go to proof.

Proposition 0.17.5 (Preimage Distributes over Union and Intersection). *Let $f : A \rightarrow B$, $S, T \subseteq B$. Then:*

$$(i) \quad f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T).$$

$$(ii) \quad f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

$$(iii) \quad f^{-1}(S^c) = (f^{-1}(S))^c.$$

Go to proof.

0.18 Real-Valued Functions

Real-Valued Functions

Definition 0.18.1 (Pointwise Sum of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \rightarrow \mathbb{R}$. The pointwise sum $f + g : X \rightarrow \mathbb{R}$ is the function defined by

$$(f + g)(x) = f(x) + g(x) \quad (x \in X).$$

Definition 0.18.2 (Pointwise Difference of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \rightarrow \mathbb{R}$. The pointwise difference $f - g : X \rightarrow \mathbb{R}$ is the function defined by

$$(f - g)(x) = f(x) - g(x) \quad (x \in X).$$

Definition 0.18.3 (Pointwise Product of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \rightarrow \mathbb{R}$. The pointwise product $fg : X \rightarrow \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \quad (x \in X).$$

Definition 0.18.4 (Pointwise Scalar Multiple of a Function). Let $X \subseteq \mathbb{R}$, let $f : X \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \rightarrow \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \quad (x \in X).$$

Definition 0.18.5 (Pointwise Absolute Value of a Function). Let $X \subseteq \mathbb{R}$ and let $f : X \rightarrow \mathbb{R}$. The pointwise absolute value $|f| : X \rightarrow \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \quad (x \in X).$$

Definition 0.18.6 (Pointwise Maximum of Two Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \rightarrow \mathbb{R}$. The pointwise maximum $\max(f, g) : X \rightarrow \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \quad (x \in X).$$

Definition 0.18.7 (Pointwise Quotient of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \rightarrow \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \rightarrow \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \quad (x \in X).$$

Definition 0.18.8 (Linear Combination of Real-Valued Functions). Let $X \subseteq \mathbb{R}$, let $f, g : X \rightarrow \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \rightarrow \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \quad (x \in X).$$

Definition 0.18.9 (Real-Linear Rule). Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Definition 0.18.10 (Bounded Above Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \quad (x \in A).$$

Definition 0.18.11 (Bounded Below Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \quad (x \in A).$$

Definition 0.18.12 (Bounded Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \quad (x \in A).$$

Definition 0.18.13 (Supremum of a Function on a Set). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Definition 0.18.14 (Infimum of a Function on a Set). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Definition 0.18.15 (Maximum Point of a Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \quad (x \in A).$$

Definition 0.18.16 (Minimum Point of a Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \quad (x \in A).$$

Definition 0.18.17 (Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \quad (x_1, x_2 \in A).$$

Definition 0.18.18 (Strictly Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \quad (x_1, x_2 \in A).$$

Definition 0.18.19 (Decreasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \quad (x_1, x_2 \in A).$$

Definition 0.18.20 (Strictly Decreasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \quad (x_1, x_2 \in A).$$

Definition 0.18.21 (Monotone Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A .

Definition 0.18.22 (Constant Function). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \quad (x_1, x_2 \in A).$$

0.19 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$

Definition 0.19.1 (ε -Neighbourhood). Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The ε -**neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Definition 0.19.2 (Deleted ε -Neighbourhood). Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε -neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Definition 0.19.3 (Cluster Point). Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0, \exists y \in X \setminus \{x\} : |y - x| < \varepsilon.$$

Theorem 0.19.4 (Sequential Characterization of Cluster Points). *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \rightarrow c$.*

Go to proof.

Definition 0.19.5 (Adherent Point). Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0 \exists y \in X (|y - x| < \varepsilon).$$

Definition 0.19.6 (Isolated Point). Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0 (V_\varepsilon(x) \cap X = \{x\}).$$

Definition 0.19.7 (Interior Point). Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0 (V_\varepsilon(x) \subseteq X).$$

Definition 0.19.8 (Boundary Point). Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0 (V_\varepsilon(x) \cap X \neq \emptyset) \wedge (V_\varepsilon(x) \cap X^c \neq \emptyset).$$

Definition 0.19.9 (Interior). Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 V_\varepsilon(x) \subseteq X\}.$$

Definition 0.19.10 (Boundary). Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0 (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Definition 0.19.11 (Closure).

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0 V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Lemma 0.19.12 (Elementary Properties of Closures). *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*

Go to proof.

Definition 0.19.13 (Closed Subset of $\mathbb{R}$). E is closed $\iff \overline{E} = E$.

Corollary 0.19.14 (Closed iff Sequentially Closed). X is closed $\iff \forall (a_n) \subseteq X, a_n \rightarrow x \Rightarrow x \in X$.

Go to proof.

Corollary 0.19.15 (Every Point of an Interval Is a Cluster Point). *If I is an interval, then every $x \in I$ is a cluster point of I .*

Go to proof.

Definition 0.19.16 (Bounded Subset of $\mathbb{R}$). X is bounded $\iff \exists M > 0, \forall x \in X : |x| \leq M$.

Theorem 0.19.17 (Heine–Borel Theorem for $\mathbb{R}$). X is closed $\wedge X$ is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X .

Go to proof.

Definition 0.19.18 (True Near a Point). Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x))$.

Proofs

Discrete Calculus

0.20 Motivation

Motivation

Discrete Analogs of Calculus

Telescoping Phenomena

Why Discrete Calculus Matters

A Structural View

0.21 Foundations

0.21.1 Foundations

0.21.2 Summation Notation

Basic Properties of Summation

Splitting Sums

Changing the Index

Empty Sums

Nested Summations

Common Examples

0.21.3 The Binomial Theorem

Theorem 0.21.1 (Binomial Theorem). *For any integer $n \geq 0$ and real numbers x, y ,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k,$$

where the binomial coefficients are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Basic Properties of Binomial Coefficients

Connection with Discrete Calculus

Proposition 0.21.2 (Higher difference formula). *For any function f ,*

$$\Delta^n f(x) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Explicit Examples

0.22 Algebra of Summations

Linearity

Splitting a Sum

Index Renaming

Index Shifting

Constant Sum

Empty Sum

Nested Sums

Common Summation Identities

0.23 Difference Operators

0.23.1 Difference Operators

Motivation

Forward Difference

Definition 0.23.1 (Forward Difference Operator). Let $f : \mathbb{Z} \rightarrow \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Example

0.23.2 Higher Differences

Definition 0.23.2 (Higher Differences). The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k -th difference $\Delta^k f(n)$ denotes the k -fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \quad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Example

0.23.3 Properties of the Difference Operator

Linearity

Proposition 0.23.3 (Linearity of Δ). For functions $f, g : \mathbb{Z} \rightarrow \mathbb{R}$ and constants $a, b \in \mathbb{R}$,

$$\Delta(af + bg) = a\Delta f + b\Delta g.$$

Constant Rule

Proposition 0.23.4 (Constant rule). If c is a constant function, then $\Delta c = 0$.

The Shift Operator and Operator Algebra

Definition 0.23.5 (Shift Operator). The *shift operator* E and the *identity operator* I are defined by

$$Ef(n) = f(n+1), \quad If(n) = f(n).$$

Theorem 0.23.6 ($\Delta = E - I$).

$$\Delta = E - I.$$

0.23.4 Difference Tables and Polynomial Detection

Definition 0.23.7 (Difference Table). Given a sequence $f(0), f(1), f(2), \dots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Example

Theorem 0.23.8 (Polynomial detection). Let $f(n)$ be a sequence generated by a polynomial of degree d . Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.

0.24 Telescoping Sums

0.24.1 Telescoping Sums

A Simple Example

General Form

The Telescoping Identity

Cancellation Pattern

Operator Form

0.25 Discrete Antiderivatives

0.25.1 Discrete Antiderivatives

Definition 0.25.1 (Discrete Antiderivative). Let $f : \mathbb{Z} \rightarrow \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \rightarrow \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Examples

Constant of Integration

Proposition 0.25.2 (Constant of integration). *If F is a discrete antiderivative of f , then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Fundamental Theorem of Finite Calculus

0.26 Polynomial Basis

0.26.1 Polynomial Basis: Falling Factorials

The Difference Rule for Falling Factorials

Summary Table

Polynomial Representation in the Falling Factorial Basis

Consequences of the Difference Rule

Higher Differences of Falling Factorials

Connection to Binomial Coefficients

Worked Example: Expanding n^2

0.27 Calculus Rules

0.27.1 Calculus Rules

Product Rule

Summation by Parts

0.28 Expansions

0.28.1 Newton's Forward Difference Expansion

Structural Parallel with the Taylor Series

Proof Sketch

Reading Coefficients from the Difference Table

Worked Example: Expanding $f(n) = n^3$

Application: Computing $\sum_{k=0}^{n-1} k^2$

0.29 Special Functions

0.29.1 Special Functions in Discrete Calculus

The Discrete Exponential

Definition 0.29.1 (Discrete Exponential). For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1 + a)$ is the function

$$E_a(n) = (1 + a)^n.$$

Proposition 0.29.2 (Discrete exponential rule).

$$\Delta(1 + a)^n = a(1 + a)^n.$$

0.29.2 The Function 2^n in Discrete Calculus

Differencing 2^n

Antiderivative and Geometric Sum

Higher Differences of 2^n

Proposition 0.29.3 (Higher differences of 2^n). *For all $k \geq 0$,*

$$\Delta^k 2^n = 2^n.$$

2^n in Combinatorics and the Newton Expansion

Discrete Differential Equation

Proposition 0.29.4 (Discrete IVP). *The unique function $F : \mathbb{N} \rightarrow \mathbb{R}$ satisfying*

$$F(n+1) = 2F(n), \quad F(0) = 1$$

is $F(n) = 2^n$.

Difference Formulas for Special Functions

Antiderivative Table

Proofs

Sequences

0.30 Sequence Definitions

0.30.1 Definitions

Definition 0.30.1 (Sequence). Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \rightarrow X.$$

The value $x(n)$ is written x_n and called the **n -th term** of the sequence. The sequence itself is denoted $(x_n)_{n \in \mathbb{N}}$, or simply (x_n) when the index set is clear.

Definition 0.30.2 (Real Sequence). A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \rightarrow \mathbb{R}.$$

0.30.2 Null and Constant Sequences

Null and Constant Sequences

Definition 0.30.3 (Constant Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Definition 0.30.4 (Null Sequence). Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Theorem 0.30.5 (Constant Sequence Convergence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c .*

Go to proof.

Theorem 0.30.6 (Zero Sequence is Null). *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.7 (Constant Null Sequence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Definition 0.30.8 (Ultimately Constant Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Theorem 0.30.9 (Difference from the Limit is Null). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Go to proof.

Theorem 0.30.10 (Ultimately Constant Sequence Convergence). *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c .*

Go to proof.

Theorem 0.30.11 (Constant Implies Ultimately Constant). *Every constant real sequence is ultimately constant.*

Go to proof.

Theorem 0.30.12 (Ultimately Zero Sequence is Null). *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.13 (Ultimately Constant Null Sequence). *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Theorem 0.30.14 (Tail Equality Preserves Convergence). *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L .*

Go to proof.

Definition 0.30.15 (Sequence Bounded Above). *Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.*

Definition 0.30.16 (Sequence Bounded Below). *Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.*

Definition 0.30.17 (Bounded Sequence). *Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.*

Theorem 0.30.18 (Eventually Bounded Above Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.19 (Eventually Bounded Below Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.20 (Eventually Bounded Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.21 (Bounded Sequence If and Only If Bounded Above and Below). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Theorem 0.30.22 (Absolute Bound Gives Upper and Lower Bounds). *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \leq K$ for every $n \in \mathbb{N}$, then $-K \leq x_n \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

Theorem 0.30.23 (Upper and Lower Bounds Give an Absolute Bound). *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

0.31 Examples and Counterexamples

Examples and Counterexamples

0.32 Convergence

Definition 0.32.1 (Convergence of a Sequence). Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon).$$

The value L is called the **limit** of (x_n) , written $x_n \rightarrow L$ or $\lim_{n \rightarrow \infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Definition 0.32.2 (Convergence — Neighborhood Formulation). Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε -neighborhood

$$V_\varepsilon(L) := (L - \varepsilon, L + \varepsilon)$$

contains all but finitely many terms of (x_n) ; that is,

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (x_n \in V_\varepsilon(L)).$$

Theorem 0.32.3 (Equivalence of Convergence Formulations). *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

- (a) $x_n \rightarrow L$.
- (b) $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (|x_n - L| < \varepsilon)$.
- (c) $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (L - \varepsilon < x_n < L + \varepsilon)$.
- (d) $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (x_n \in V_\varepsilon(L))$.

Go to proof.

0.32.1 Limits

Theorem 0.32.4 (Uniqueness of Limits). *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \rightarrow L$ and $x_n \rightarrow K$, then $L = K$.*

Go to proof.

Theorem 0.32.5 (Limit Preserves Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \rightarrow L$ and $y_n \rightarrow M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*

Go to proof.

Theorem 0.32.6 (Strict Limit Separation Gives Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \rightarrow A$ and $y_n \rightarrow B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.7 (Eventual Strict Comparison Preserves Weak Limit Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \rightarrow A$ and $y_n \rightarrow B$.*

- (a) *If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.*
- (b) *If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.*

Go to proof.

Theorem 0.32.8 (Constant Comparison for Sequence Limits). *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \rightarrow A$.*

- (a) *If there exists $N \in \mathbb{N}$ such that $x_n \leq B$ for every $n \geq N$, then $A \leq B$.*
- (b) *If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \geq N$, then $A \leq B$.*
- (c) *If there exists $N \in \mathbb{N}$ such that $x_n \geq B$ for every $n \geq N$, then $A \geq B$.*
- (d) *If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \geq N$, then $A \geq B$.*

Go to proof.

Theorem 0.32.9 (Constant Squeeze Theorem). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \leq x_n \leq L$ for every $n \geq N_0$, then $x_n \rightarrow L$.*

Go to proof.

Theorem 0.32.10 (Sequence Squeeze Theorem). *Let (a_n) , (x_n) , and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \rightarrow L$ and $b_n \rightarrow L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \leq x_n \leq b_n$ for every $n \geq N_0$, then $x_n \rightarrow L$.*

Go to proof.

Theorem 0.32.11 (Absolute-Value Squeeze Theorem). *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \rightarrow 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \leq u_n$ for every $n \geq N_0$, then $x_n \rightarrow L$.*

Go to proof.

0.32.2 Algebra of Limits

Algebra of Convergent Sequences

Definition 0.32.12 (Pointwise Sum of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x + y)_n := x_n + y_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.13 (Pointwise Difference of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x - y)_n := x_n - y_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.14 (Scalar Multiple of a Sequence). Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.15 (Linear Combination of Real Sequences). Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.16 (Pointwise Negation of a Sequence). Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.17 (Pointwise Product of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.18 (Reciprocal Sequence). Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.19 (Pointwise Quotient of Sequences). Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.20 (Square Sequence). Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.21 (Absolute Value Sequence). Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.22 (Square Root Sequence). Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.32.23 (Limit of a Scalar Multiple). Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \rightarrow L$, then $\alpha x_n \rightarrow \alpha L$.

Go to proof.

Theorem 0.32.24 (Limit of a Sum). Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \rightarrow L$ and $y_n \rightarrow M$, then $x_n + y_n \rightarrow L + M$.

Go to proof.

Theorem 0.32.25 (Limit of a Negation). Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then $-x_n \rightarrow -L$.

Go to proof.

Theorem 0.32.26 (Limit of a Difference). Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \rightarrow L$ and $y_n \rightarrow M$, then $x_n - y_n \rightarrow L - M$.

Go to proof.

Theorem 0.32.27 (Limit of a Product). Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \rightarrow L$ and $y_n \rightarrow M$, then $x_n y_n \rightarrow LM$.

Go to proof.

Theorem 0.32.28 (Nonzero Limit is Eventually Nonzero). Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \rightarrow L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.

Go to proof.

Theorem 0.32.29 (Limit of a Reciprocal). Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \rightarrow L$, then $1/x_n \rightarrow 1/L$.

Go to proof.

Theorem 0.32.30 (Limit of a Quotient). Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \rightarrow L$ and $y_n \rightarrow M$, then $x_n/y_n \rightarrow L/M$.

Go to proof.

Theorem 0.32.31 (Limit of a Square). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then $x_n^2 \rightarrow L^2$.*

Go to proof.

Theorem 0.32.32 (Limit of an Absolute Value). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then $|x_n| \rightarrow |L|$.*

Go to proof.

Theorem 0.32.33 (Positive Limit is Eventually Positive). *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \rightarrow L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.34 (Limit of a Square Root). *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then $0 \leq L$ and $\sqrt{x_n} \rightarrow \sqrt{L}$.*

Go to proof.

Theorem 0.32.35 (Polynomial Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \rightarrow L$, then $p(x_n) \rightarrow p(L)$.*

Go to proof.

Theorem 0.32.36 (Rational Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \rightarrow L$, then*

$$\frac{p(x_n)}{q(x_n)} \rightarrow \frac{p(L)}{q(L)}.$$

Go to proof.

0.33 Tails

Definition 0.33.1 (M -Tail of a Sequence). *Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The M -tail of (x_n) is the sequence*

$$(x_n)_{n \geq M} := (x_M, x_{M+1}, x_{M+2}, \dots).$$

Theorem 0.33.2 (Convergence of a Tail). *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m -tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \rightarrow \infty} x_{m+n} = \lim_{n \rightarrow \infty} x_n.$$

Go to proof.

Theorem 0.33.3 (Convergence by Domination). *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n \rightarrow \infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \leq c a_n \quad \forall n \geq m,$$

then $\lim_{n \rightarrow \infty} x_n = L$.

Go to proof.

Theorem 0.33.4 (Ratio Limit Less Than One Implies Null). *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n \rightarrow \infty} x_n = 0.$$

Go to proof.

0.34 Monotonicity

Monotonicity of Sequences

Definition 0.34.1 (Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \leq x_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.2 (Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \leq x_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.3 (Strictly Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.4 (Strictly Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.5 (Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Definition 0.34.6 (Eventually Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \leq x_{n+1} \quad \text{for every } n \geq N.$$

Definition 0.34.7 (Eventually Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \quad \text{for every } n \geq N.$$

Definition 0.34.8 (Eventually Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Theorem 0.34.9 (Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) *If (x_n) is increasing and bounded above, then (x_n) converges and*

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) *If (x_n) is decreasing and bounded below, then (x_n) converges and*

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.34.10 (Strict Monotonicity Implies Monotonicity). *Let (x_n) be a real sequence.*

(a) *If (x_n) is strictly increasing, then (x_n) is increasing.*

(b) *If (x_n) is strictly decreasing, then (x_n) is decreasing.*

Go to proof.

Theorem 0.34.11 (Bounded Monotone Sequence Equivalences). *Let (x_n) be a real sequence.*

(a) *If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.*

- (b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.
- (c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Theorem 0.34.12 (Eventually Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

- (a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.
- (b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.
- (c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Theorem 0.34.13 (Unbounded Monotone Divergence). *Let (x_n) be a real sequence.*

- (a) If (x_n) is increasing and is not bounded above, then $x_n \rightarrow +\infty$.
- (b) If (x_n) is decreasing and is not bounded below, then $x_n \rightarrow -\infty$.

Go to proof.

Theorem 0.34.14 (Algebraic Transformations Preserve Monotonicity). *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

- (a) If (x_n) is increasing, then $(x_n + c)$ is increasing.
- (b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.
- (c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.
- (d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.
- (e) If (x_n) is increasing, then $(-x_n)$ is decreasing.
- (f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

0.35 Subsequences

Subsequences

Definition 0.35.1 (Strictly Increasing Index Map). Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \implies \sigma(k) < \sigma(\ell) \quad \text{for all } k, \ell \in \mathbb{N}.$$

Definition 0.35.2 (Subsequence). Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k \in \mathbb{N}}$, where $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ is a strictly increasing index map.

Definition 0.35.3 (Subsequential Limit). Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k \rightarrow \infty} x_{\sigma(k)} = L.$$

Definition 0.35.4 (Has Convergent Subsequence). Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ and a real number L such that

$$\lim_{k \rightarrow \infty} x_{\sigma(k)} = L.$$

Theorem 0.35.5 (Subsequence Indices Dominate the Identity). Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing index map. Then

$$k \leq \sigma(k) \quad \text{for every } k \in \mathbb{N}.$$

Go to proof.

Theorem 0.35.6 (Subsequences Preserve Limits). Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then every subsequence of (x_n) converges to L .

Go to proof.

Theorem 0.35.7 (Subsequential Limit of a Convergent Sequence). Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \rightarrow L$, then L is the only possible subsequential limit of (x_n) .

Go to proof.

Theorem 0.35.8 (Divergence by Two Subsequential Limits). Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.

Go to proof.

Theorem 0.35.9 (Boundedness Passes to Subsequences). Every subsequence of a bounded real sequence is bounded.

Go to proof.

Theorem 0.35.10 (Monotonicity Passes to Subsequences). Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.

Go to proof.

Theorem 0.35.11 (Subsequence of a Subsequence). Every subsequence of a subsequence of (x_n) is a subsequence of (x_n) .

Go to proof.

Theorem 0.35.12 (Eventual Properties Pass to Subsequences). Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.

Go to proof.

Theorem 0.35.13 (Frequent Properties Yield Subsequences). Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.

Go to proof.

Theorem 0.35.14 (Subsequential Limits Respect Bounds). Let (x_n) be a real sequence and let L be a subsequential limit of (x_n) . If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.

Go to proof.

Theorem 0.35.15 (Squeeze Passes to Subsequences). Let (a_n) , (x_n) , and (b_n) be real sequences. If $a_n \leq x_n \leq b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \quad \text{for every } k \geq K.$$

Go to proof.

Theorem 0.35.16 (Monotone Subsequence Theorem). *Every real sequence has a monotone subsequence.*

Go to proof.

Theorem 0.35.17 (Bolzano-Weierstrass Theorem for Sequences). *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Theorem 0.35.18 (Sequential Compactness of Closed Bounded Intervals). *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

0.36 Divergence

Divergence

Definition 0.36.1 (Divergent Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \rightarrow L$.

Definition 0.36.2 (Divergence to Positive Infinity). Let (x_n) be a real sequence. The sequence (x_n) **diverges to $+\infty$** if

$$\forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n).$$

Definition 0.36.3 (Divergence to Negative Infinity). Let (x_n) be a real sequence. The sequence (x_n) **diverges to $-\infty$** if

$$\forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M).$$

Definition 0.36.4 (Oscillatory Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Theorem 0.36.5 (Divergence to Infinity Implies Real Divergence). *Let (x_n) be a real sequence. If $x_n \rightarrow +\infty$ or $x_n \rightarrow -\infty$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.6 (Two Subsequential Limits Force Divergence). *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \rightarrow +\infty$.*

Go to proof.

Theorem 0.36.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \rightarrow -\infty$.*

Go to proof.

Theorem 0.36.9 (Bounded Divergence Produces Two Subsequential Limits). *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n) .*

Go to proof.

0.37 Liminf and Limsup

Liminf and Limsup

Definition 0.37.1 (Tail Supremum Sequence). Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.2 (Tail Infimum Sequence). Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \quad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.3 (Limit Superior of a Sequence). Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} s_n.$$

Definition 0.37.4 (Limit Inferior of a Sequence). Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} i_n.$$

Theorem 0.37.5 (Tail Suprema Are Decreasing). *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Theorem 0.37.6 (Tail Infima Are Increasing). *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Theorem 0.37.7 (Liminf Is Below Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

Go to proof.

Theorem 0.37.8 (Convergence iff Liminf Equals Limsup). *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \rightarrow L$ if and only if*

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L.$$

Go to proof.

Theorem 0.37.9 (Limsup Is the Largest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n \rightarrow \infty} x_n$. Then S is a subsequential limit of (x_n) , and every subsequential limit of (x_n) is less than or equal to S .*

Go to proof.

Theorem 0.37.10 (Liminf Is the Smallest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n \rightarrow \infty} x_n$. Then I is a subsequential limit of (x_n) , and every subsequential limit of (x_n) is greater than or equal to I .*

Go to proof.

Theorem 0.37.11 (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n < \limsup_{n \rightarrow \infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Theorem 0.37.12 (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n.$$

Go to proof.

Theorem 0.37.13 (Liminf Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n.$$

Go to proof.

Theorem 0.37.14 (Limsup Squeeze Under Eventual Order). *Let (a_n) , (x_n) , and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

then

$$\limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} b_n.$$

Go to proof.

Theorem 0.37.15 (Liminf Squeeze Under Eventual Order). *Let (a_n) , (x_n) , and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

then

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} b_n.$$

Go to proof.

0.38 Order-Theoretic Limit Constructions

Order-Theoretic Limit Constructions

Theorem 0.38.1 (Increasing Sequence Limit as Supremum). *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.2 (Decreasing Sequence Limit as Infimum). *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.3 (Tail Suprema and Infima Converge). *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \quad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Theorem 0.38.4 (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

0.39 Cluster Values of Sequences

Cluster Values of Sequences

Definition 0.39.1 (Cluster Value of a Sequence). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that*

$$|x_n - L| < \varepsilon.$$

Theorem 0.39.2 (Cluster Values Are Subsequential Limits). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n) .*

Go to proof.

Theorem 0.39.3 (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*

Go to proof.

Theorem 0.39.4 (Limsup and Liminf Are Extremal Cluster Values). *Let (x_n) be a bounded real sequence. Then $\limsup_{n \rightarrow \infty} x_n$ is the largest cluster value of (x_n) , and $\liminf_{n \rightarrow \infty} x_n$ is the smallest cluster value of (x_n) .*

Go to proof.

0.40 Cauchy

Cauchy Sequences

Definition 0.40.1 (Cauchy Sequence). *Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if*

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon).$$

Theorem 0.40.2 (Convergent Sequences Are Cauchy). *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.3 (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

Go to proof.

Theorem 0.40.4 (Cauchy Sequence with a Convergent Subsequence). *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L .*

Go to proof.

Theorem 0.40.5 (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.6 (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N -tail has pairwise term distances less than ε .*

Go to proof.

Theorem 0.40.7 (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Theorem 0.40.8 (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \rightarrow 0.$$

Go to proof.

Theorem 0.40.9 (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Theorem 0.40.10 (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.11 (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.12 (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.13 (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.14 (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.15 (Quotients of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*

Go to proof.

Theorem 0.40.16 (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

0.41 Applications of Sequences

Applications of Sequences

Definition 0.41.1 (Newton Sequence for $\sqrt{2}$). The **Newton sequence for $\sqrt{2}$** is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \quad x_{n+1} := \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.2 (Newton Approximation of $\sqrt{2}$). *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n \rightarrow \infty} x_n = \sqrt{2}.$$

Go to proof.

Definition 0.41.3 (Factorial Partial Sums). The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^n \frac{1}{k!} \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.4 (Factorial Partial Sums Approximate e). *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e :*

$$\lim_{n \rightarrow \infty} s_n = e.$$

Go to proof.

Definition 0.41.5 (Compound-Interest Sequence). The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n} \right)^n \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.6 (Compound-Interest Approximation of e). *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n \rightarrow \infty} c_n = e.$$

Go to proof.

Definition 0.41.7 (Decimal Truncation Sequence). Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.8 (Decimal Truncations Converge). *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α . Then*

$$\lim_{n \rightarrow \infty} d_n = \alpha.$$

Go to proof.

Proofs

Capstone

Series

0.42 Finite Series

Definition 0.42.1 (Finite Real Sequence on an Integer Interval). Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^n$.

Definition 0.42.2 (Empty Finite Sum). Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^n a_i := 0.$$

Definition 0.42.3 (Finite Sum). Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^n a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n .

Definition 0.42.4 (Recursive Finite Sum). Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^n a_i \right) + a_{n+1} \quad \text{whenever } n \geq m - 1.$$

Lemma 0.42.5 (Splitting a Finite Sum). Let $m \leq n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq p$. Then

$$\sum_{i=m}^n a_i + \sum_{i=n+1}^p a_i = \sum_{i=m}^p a_i.$$

Go to proof.

Definition 0.42.6 (Summation over a Finite Set). Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \rightarrow \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^n f(g(i)).$$

Proposition 0.42.7 (Finite Summations Are Well-Defined). Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \rightarrow \mathbb{R}$, and let

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X \quad \text{and} \quad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X$$

be bijections. Then

$$\sum_{i=1}^n f(g(i)) = \sum_{i=1}^n f(h(i)).$$

Go to proof.

Proposition 0.42.8 (Empty Finite Set Summation). *If X is empty, and $f : X \rightarrow \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Proposition 0.42.9 (Singleton Finite Set Summation). *If X consists of a single element, $X = \{x_0\}$, and $f : X \rightarrow \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Proposition 0.42.10 (Substitution for Finite Set Summation). *If X is a finite set, $f : X \rightarrow \mathbb{R}$ is a function, and $g : Y \rightarrow X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Proposition 0.42.11 (Integer Interval as a Finite Set). *Let $n \leq m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^m a_i = \sum_{i \in X} a_i.$$

Go to proof.

Proposition 0.42.12 (Disjoint Union Finite Set Summation). *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \rightarrow \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Proposition 0.42.13 (Addition Rule for Finite Set Summation). *Let X be a finite set, and let $f : X \rightarrow \mathbb{R}$ and $g : X \rightarrow \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.14 (Scalar Multiplication Rule for Finite Set Summation). *Let X be a finite set, let $f : X \rightarrow \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Proposition 0.42.15 (Monotonicity for Finite Set Summation). *Let X be a finite set, and let $f : X \rightarrow \mathbb{R}$ and $g : X \rightarrow \mathbb{R}$ be functions such that $f(x) \leq g(x)$ for all $x \in X$. Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.16 (Triangle Inequality for Finite Set Summation). *Let X be a finite set, and let $f : X \rightarrow \mathbb{R}$ be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

Go to proof.

Lemma 0.42.17 (Iterated Summation over a Product). *Let X and Y be finite sets, and let $f : X \times Y \rightarrow \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y).$$

Go to proof.

Corollary 0.42.18 (Fubini's Theorem for Finite Series). *Let X and Y be finite sets, and let $f : X \times Y \rightarrow \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y) = \sum_{(y, x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left(\sum_{x \in X} f(x, y) \right).$$

Go to proof.

Lemma 0.42.19 (Reindexing a Finite Sum). *Let $m \leq n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^n a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Lemma 0.42.20 (Finite Sum of Sums). *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^n (a_i + b_i) = \left(\sum_{i=m}^n a_i \right) + \left(\sum_{i=m}^n b_i \right).$$

Go to proof.

Lemma 0.42.21 (Finite Sum of Scalar Multiples). *Let $m \leq n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^n (ca_i) = c \left(\sum_{i=m}^n a_i \right).$$

Go to proof.

Lemma 0.42.22 (Triangle Inequality for Finite Series). *Let $m \leq n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i|.$$

Go to proof.

Lemma 0.42.23 (Comparison Test for Finite Series). *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Suppose that $a_i \leq b_i$ for all i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^n a_i \leq \sum_{i=m}^n b_i.$$

Go to proof.

0.43 Infinite Series

Definition 0.43.1 (Formal Infinite Series). A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \geq m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots.$$

Proofs

Function Sequences

Proofs